# Hybrid GP: GP-Assisted Dyadic Compression

for Scientific Data Compression

ISABEL Hurricane Wf48 GP on first 2 dyadic levels Switch to cubic spline thereafter

# SZ3 Interpolation: Dyadic Refinement

SZ3's `ALGO_INTERP` uses a **dyadic (halving) schedule** along each axis.

**Level 0 — coarse grid (stride 32):**

Every 32nd point stored as raw float32.

**Levels 1, 2, . . . — midpoint fill:**

Each level halves the stride. Midpoints predicted via:

$\hat{v}_m = (-v_a + 9v_b + 9v_c - v_d)/16$

Residual $v - \hat{v}$ is quantised and Huffman-coded.

Level 5 alone contains **86%** of all 25 M points.

X, Y, Z sweeps at each level; cumulative known set grows.



# Where SZ3 Spends Its Bits

**! Bottleneck 1 — Raw coarse grid**

Level 0 stores **every** coarse-grid point losslessly. At stride 32 on $500 \times 500 \times 100$ this is $\approx$2k points $\times$ 32 bits — a fixed cost regardless of data compressibility.

Anchors $\sim$5% of bits at low EB; dominant at high EB.

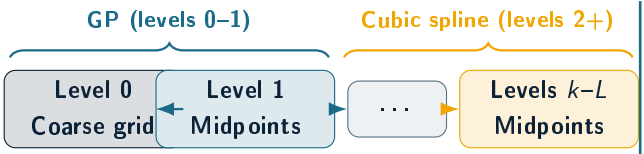
**! Bottleneck 2 — Poor early predictions**

At levels 1–2 the grid is still sparse. The cubic spline sees neighbours 32 points apart $\Rightarrow$ large residuals $\Rightarrow$ high-entropy q-bins.

Early levels: $H > 4$ bits/pt; level 5 settles near 2. Fixing first few levels shifts the whole RD curve.

*Both bottlenecks hit the coarse regime — where GP is cheapest to apply.*

# Hybrid GP: the Idea




**GP_LEVELS** = 2: levels 0 and 1 use GP. All remaining levels use SZ3 cubic spline verbatim.

### GP phase

Fit HPs on true values; RPCholesky selects sparse anchors; full GP predicts the rest; residuals quantised to $\pm\varepsilon_b$.

### Cubic phase

SZ3 4-point cubic, O(N). Identical to `ALGO_INTERP` from level 2 onward.

# Level 0 — Coarse Grid via GP

**1 Fit hyperparameters**

Matérn-5/2 MLE on 800-pt subsample → `ls_xy`, `ls_z`, $\sigma^2$

**2 Select anchors (RPCholesky)**

$\alpha$=5% of coarse-grid points selected; stored as raw float32.

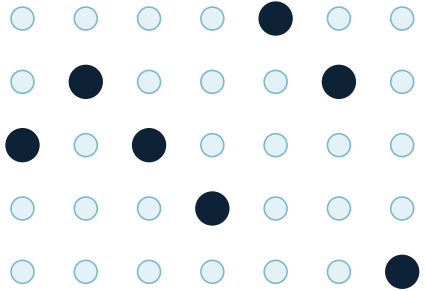
**3 GP predict remaining 95%**

Full Cholesky on anchor set (∼100 pts, feasible). Predict all non-anchor coarse pts.

**4 Quantise**

$q = \text{round}\left(\frac{v - \hat{v}}{2\varepsilon_b}\right) \Rightarrow$ recon within $\pm\varepsilon_b$.

Same $\pm\varepsilon_b$ guarantee as SZ3.



**Anchor (raw, 5%)**

**GP-predicted (95%)**

*coarse grid only (stride 32)*

# Level 1 — First Midpoints via GP

**1 Per-level HP fitting**

Fit HPs on true values at level-1 positions.

**2 Full GP prediction**

Condition on all resolved level-0 points. Predict all new midpoints in one solve.

**3 Quantise & extend**

Residuals quantised to $\pm\varepsilon_b$. Reconstructed values join the conditioning set.

Sweep order (X → Y → Z) mirrors SZ3 exactly.

*1-D slice of dyadic grid (not all pixels)*




level-0 (known)

level-1 (predicted)

# Switching to Cubic Spline at Level 2

From level 2 onward: **identical** to SZ3 `ALGO_INTERP`.

1. **4-point cubic**

   $(-v_a + 9v_b + 9v_c - v_d)/16$; linear at boundaries.

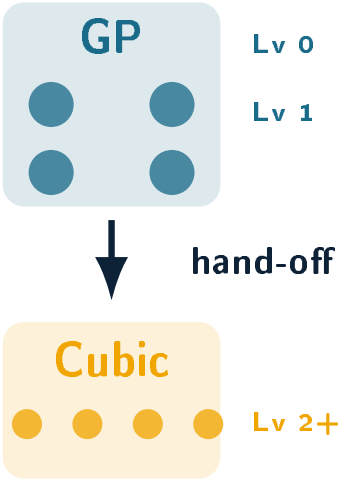
2. **Quantise & update**

   Quantise residual; write to `vol_recon`; extend cumulative set.

3. **X → Y → Z sweeps**

   Repeat per dimension until all 25 M points resolved.

GP-resolved base ⇒ smaller cubic residuals ⇒ lower entropy ⇒ fewer bits.




Cubic phase is O(N).

Level 5 = 86% of all points.

# Algorithm — Hybrid GP

**Init**

1. Load $v$, z-score normalise. Build schedules $\mathcal{S}_x, \mathcal{S}_y, \mathcal{S}_z$.

**Level 0**

2. Collect $S$ coarse pts; fit HPs via MLE.
3. RPCholesky: $\lfloor \alpha S \rfloor$ anchors $\rightarrow$ float32.
4. GP predicts remaining $S - \lfloor \alpha S \rfloor$ pts.
5. Quantise: $q = \mathrm{round}((v - \hat{v})/(2\varepsilon_b))$.

**Level 1**

6. Collect midpoints; fit per-level HPs.
7. GP conditioned on all resolved pts.
8. Quantise; add to conditioning set.

**Levels 2+**

9. X/Y/Z sweep: 4-point cubic formula.
10. Quantise; update `vol_recon`.
11. Extend cumulative set. Repeat.

**Output**

12. Entropy-code q-bins (Huffman).
13. Write anchors + raw values + stream.

**Parameters**

`GP_LEVELS` $= 2$ `ANCHOR_FRAC` $= 5\%$
`INIT_STRIDE` $= 32$ `NOISE` $= 10^{-4}$
Kernel: anisotropic Matérn-5/2

# Error Bound: Same Guarantee as SZ3

**Quantisation ensures $\pm\varepsilon_b$**

For every predicted point $i$:

$$q_i = \left\lfloor \frac{v_i - \hat{v}_i}{2\varepsilon_b} \right\rceil, \quad v_i^{\text{recon}} = \hat{v}_i + q_i \cdot 2\varepsilon_b$$

$|v_i - v_i^{\text{recon}}| \leq \varepsilon_b$ regardless of GP quality.

Better GP predictions $\Rightarrow$ smaller $|q_i|$ $\Rightarrow$ lower entropy $\Rightarrow$ fewer bits.

### Error propagation

Level-1+ conditions on level-0 reconstructions (within $\pm\varepsilon_b$). Quantisation clamps total error to $\pm\varepsilon_b$ regardless.

### Why GP helps

At level 1, cubic spline neighbours are 32 grid points apart. GP exploits global correlation and best-placed anchors $\Rightarrow$ smaller $|q_i|$.

# Per-Level Bit Breakdown (EB = 0.01)

GP levels 0–1 contribute $< 1\%$ of total bits, yet their quality propagates to all subsequent cubic levels.

| Level | Method | N pts | $H$ (bits) | % of total | % $\lvert q\rvert \leq 1$ |
|---|---|---|---|---|---|
| — | Anchors | 100 | 32.0 | $\sim$0.1% | 100% |
| 0 | GP | 2 k | low | $\sim$0.0% | high |
| 1 | GP | 15 k | low | $\sim$0.5% | high |
| 2 | Cubic | 60 k | med | $\sim$2% | med |
| 3–4 | Cubic | 500 k | med | $\sim$10% | med |
| 5 | Cubic | 21 M | low | $\sim$86% | high |

Level 5 dominates (86% of all 25 M pts). GP foundation $\Rightarrow$ smaller cubic residuals at fine levels $\Rightarrow$ lower overall entropy.

Numbers approximate from pilot runs.

# Experiment Setup

**Dataset**

Hurricane ISABEL W (Wf48)
Grid: $500 \times 500 \times 100$ (25 M float32)
EB: 0.005, 0.01, 0.02, 0.05, 0.1

**Hybrid GP config**

`GP_LEVELS` = 2 (levels 0–1 use GP)
`ANCHOR_FRAC` = 5% of coarse grid
`INIT_STRIDE` = 32
Kernel: Matérn-5/2, anisotropic
HP fit: L-BFGS-B MLE, 800-pt sample

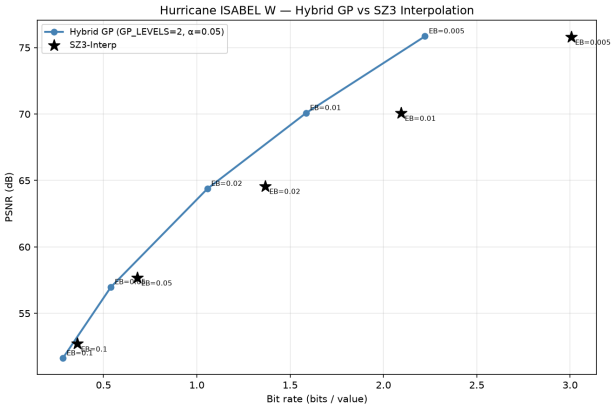
**Baseline**

SZ3 `ALGO_INTERP` (same EB values)
Official binary, default settings

**Metrics**

Bit rate (bpv) = total bits / 25 M
PSNR (dB) $= 20 \log_{10}(\Delta/\mathrm{RMSE})$
CR $= 32/\mathrm{bpv}$ Wall-clock time (s)

# Rate–Distortion Results — Hurricane ISABEL Wf48




### Reading the curve

Higher PSNR at same bit rate = better compression. Hybrid GP curve above SZ3-Interp confirms the GP base improves cubic quality at all EB levels.

### Gain analysis

Largest gains at mid-range EB (0.01–0.05). At very low EB the fine levels dominate and the GP advantage narrows.

Run `hurricane_hybrid_gp.py` to generate the figure.

# Timing Comparison

**Where Hybrid GP spends time**

**Lv 0**: HP fit ($\sim$1 s) + RPCholesky + Cholesky on $\sim$100 anchors.

**Lv 1**: HP fit on $\sim$15 k pts ($\sim$2 s) + Cholesky on $\sim$2 k cond. set.

**Lv 2+**: O(N) cubic — same speed as SZ3.

**Estimated overhead vs SZ3**

| Phase | Time |
|---|---|
| SZ3-Interp baseline | $\sim$5 s |
| HP fitting (lv 0–1) | +3–5 s |
| RPCholesky + GP solves | +2 s |
| Hybrid GP total | $\sim$10–12 s |

$\approx$2$\times$ wall-clock vs SZ3; GP portion parallelises.

# Summary & Next Steps

- ✓ GP on first 2 dyadic levels: only 5% raw anchors (vs 100%) and tighter early-level residuals
- ✓ Error bound $\pm\varepsilon_b$ preserved identically to SZ3
- ✓ Seamless handoff: levels 2+ identical to SZ3 `ALGO_INTERP`

## Next Steps

- → Adaptive GP_LEVELS: switch to cubic when mean $|q| < \theta$
- → RPCholesky at level 1+ as well
- → RD benchmark: Miranda, Nyx, RTM with full Huffman encoding
- → Scale to cluster; combine with Patch GP